

Computer Networks

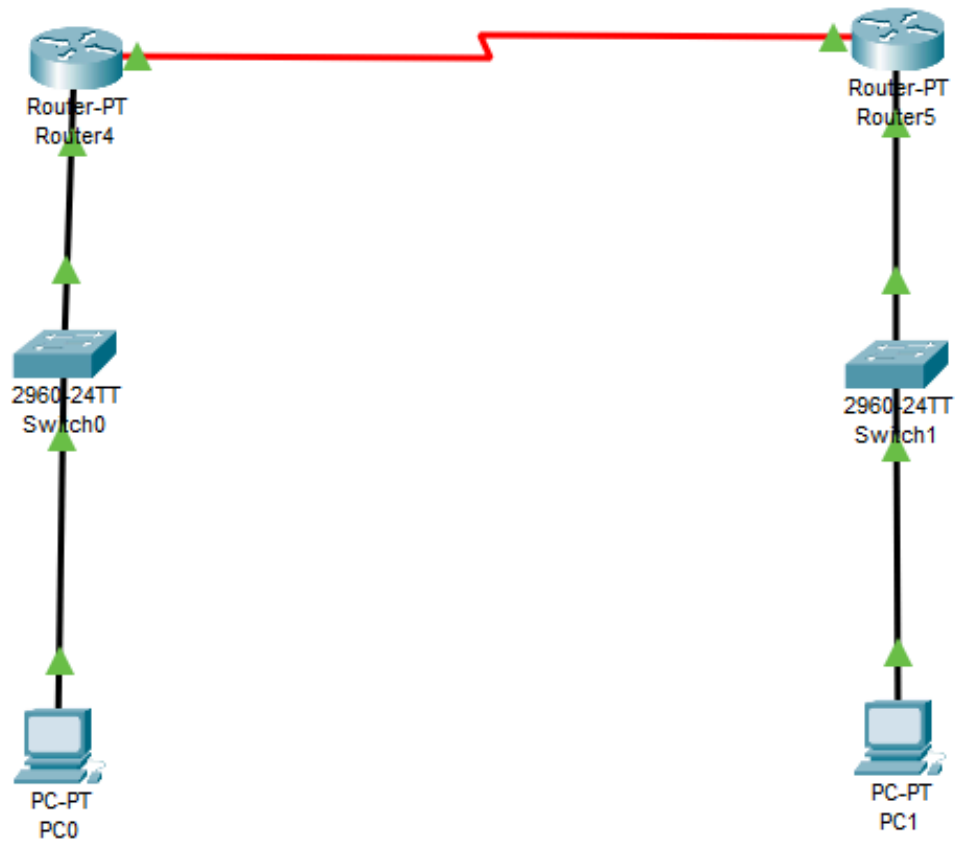
LAB#09



23K-2001

BCS-5J

Q1:



Router4

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.C71B.8755

IP Configuration

IPv4 Address 30.20.0.33

Subnet Mask 255.255.255.224

Tx Ring Limit 10

Router4

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status

☒ On

Duplex

☒ Full Duplex

Clock Rate

2000000

IP Configuration

IPv4 Address

30.20.0.65

Subnet Mask

255.255.255.224

Tx Ring Limit

10

Router5

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps☐ 10 Mbps

Duplex

☐ Half Duplex☒ Full Duplex

MAC Address

0001.971C.8490

IP Configuration

IPv4 Address

30.20.0.97

Subnet Mask

255.255.255.224

Tx Ring Limit

10

Router5

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status

☒ On

Duplex

☐ Full Duplex

Clock Rate

2000000

IP Configuration

IPv4 Address

30.20.0.66

Subnet Mask

255.255.255.224

Tx Ring Limit

10

PC0

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Bandwidth

☐ 100 Mbps☐ 10 Mbps☒ Auto

Duplex

☐ Half Duplex☒ Full Duplex☒ Auto

MAC Address

0040.0B0C.ADAC

IP Configuration

☐ DHCP☒ Static

IPv4 Address

30.20.0.34

Subnet Mask

255.255.255.224

IPv6 Configuration

☐ Automatic☒ Static

IPv6 Address

Link Local Address

FE80::240:BFF:FE0C:ADAC

PC1

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Bandwidth

☒ 100 Mbps☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex☒ Full Duplex

☒ Auto

MAC Address000B.BE48.4392

IP Configuration

☐ DHCP

☒ Static

IPv4 Address30.20.0.98

Subnet Mask255.255.255.224

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address:FE80::20B:BEFF:FE48:4392

Router4

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

RIP Routing (v2)

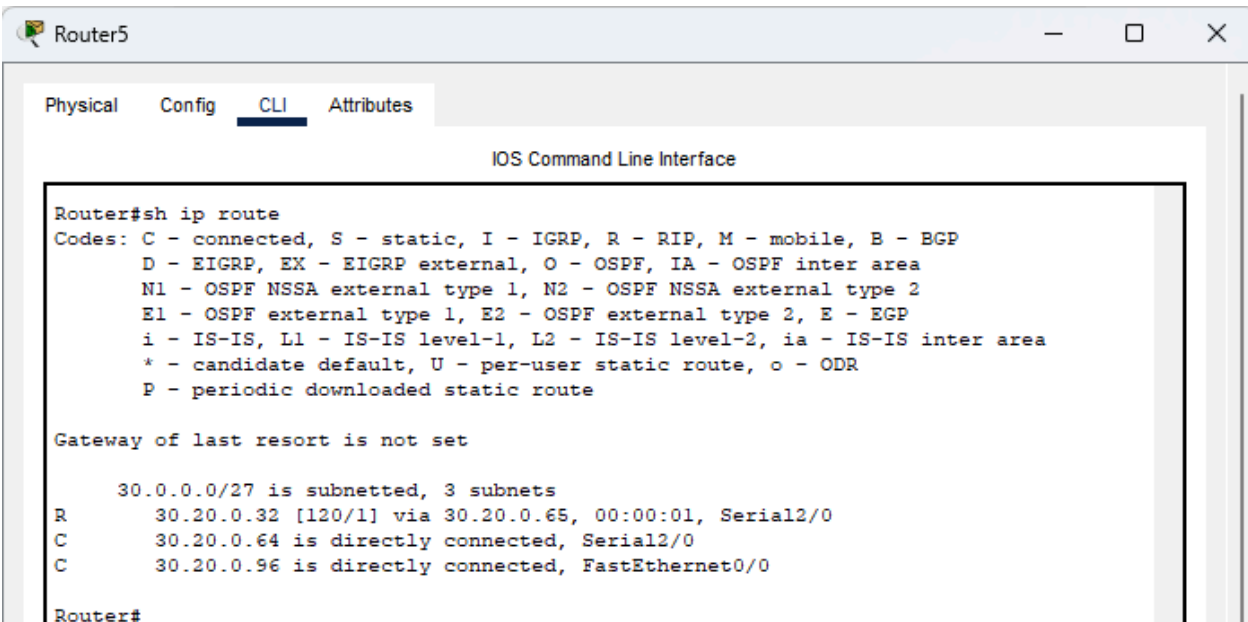
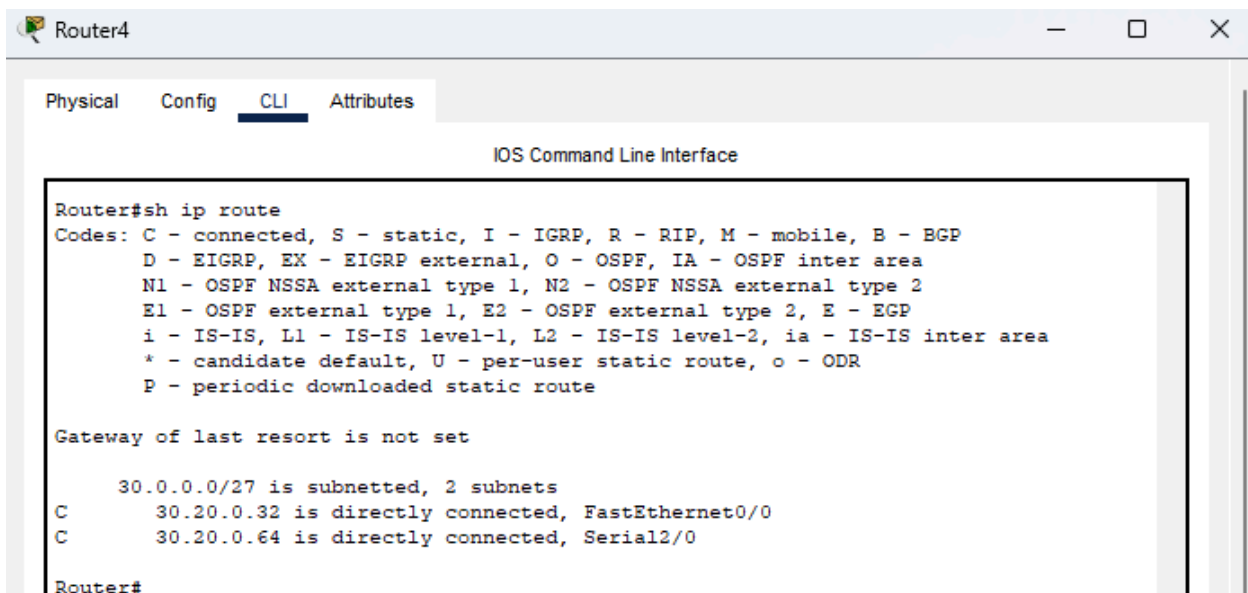
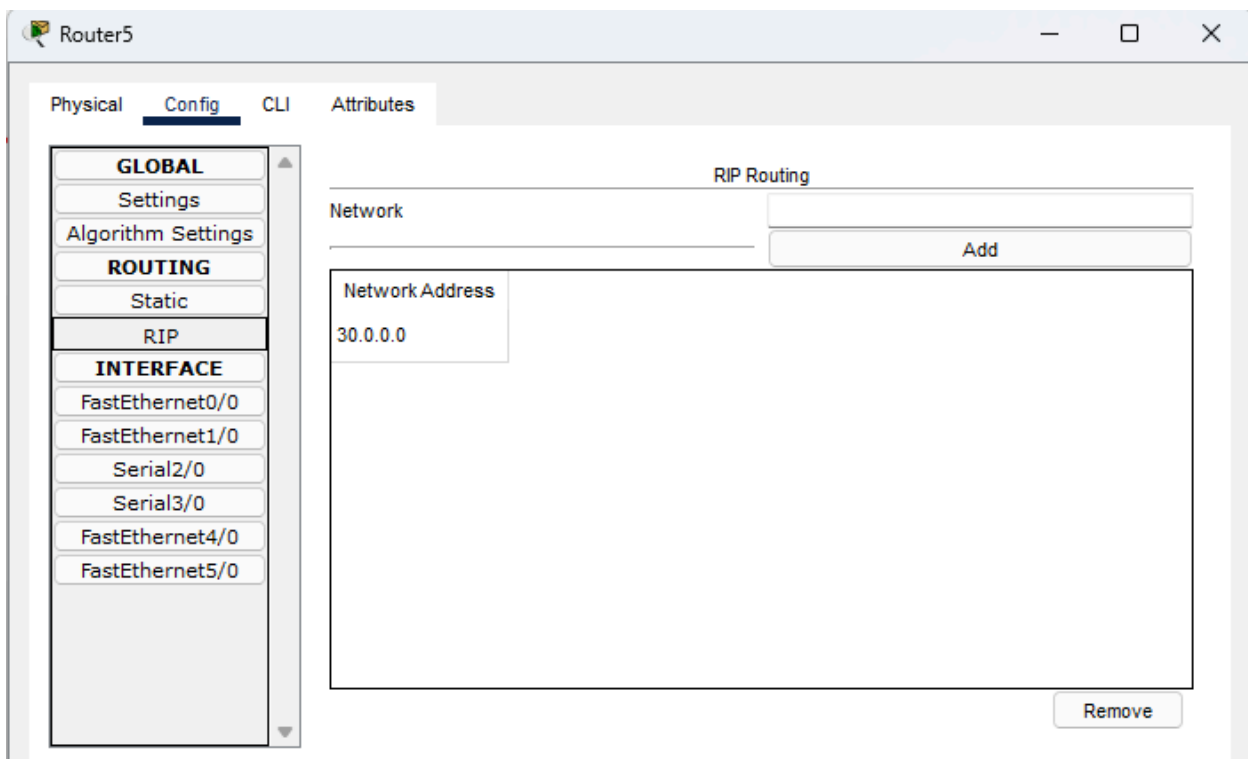
Network

Add

Network Address

30.0.0.0

Remove



Q2:

Question 2:-

Academics = 120

Development = 35

Accounts = 10

Academics

Hosts = 120 nearest power of 2 = 128

Host bits = 7 subnet mask = /25

Subnet address = 192.168.50.0/25

~~Sub~~ Range = 192.168.50.1 - 192.168.50.126

Broadcast address = 192.168.50.127

Development

Hosts = 35 nearest power of 2 = 64

Host bits = 6 subnet mask = /26

Subnet address = 192.168.50.128/26

Broadcast address = ~~192~~ 192.168.50.191

Range = 192.168.50.129 - 192.168.50.190

Accounts

Hosts = 10 nearest power of 2 = 16

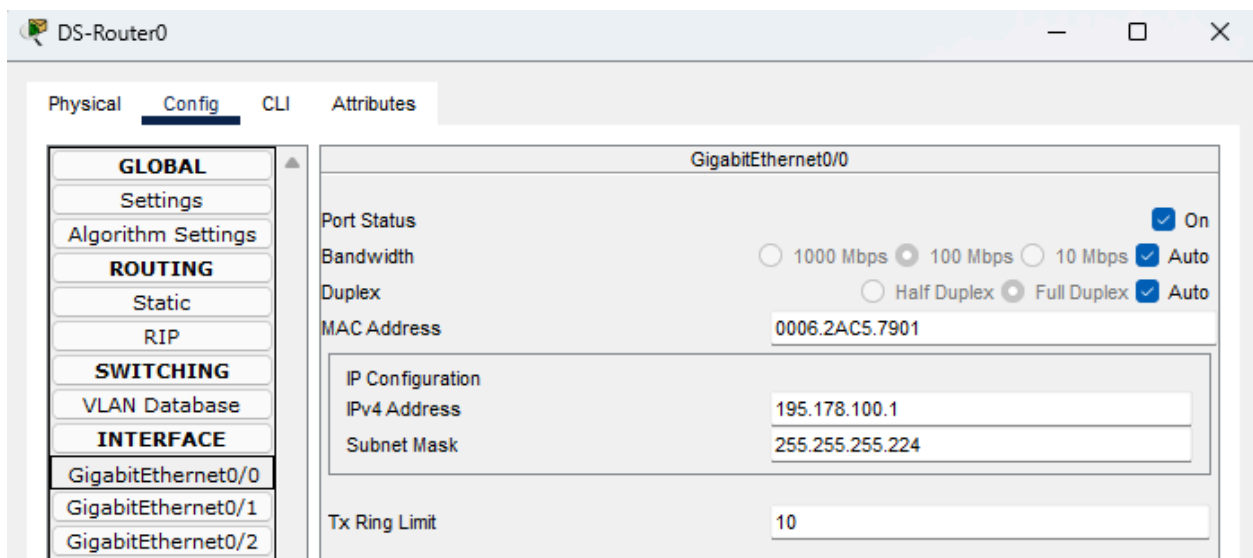
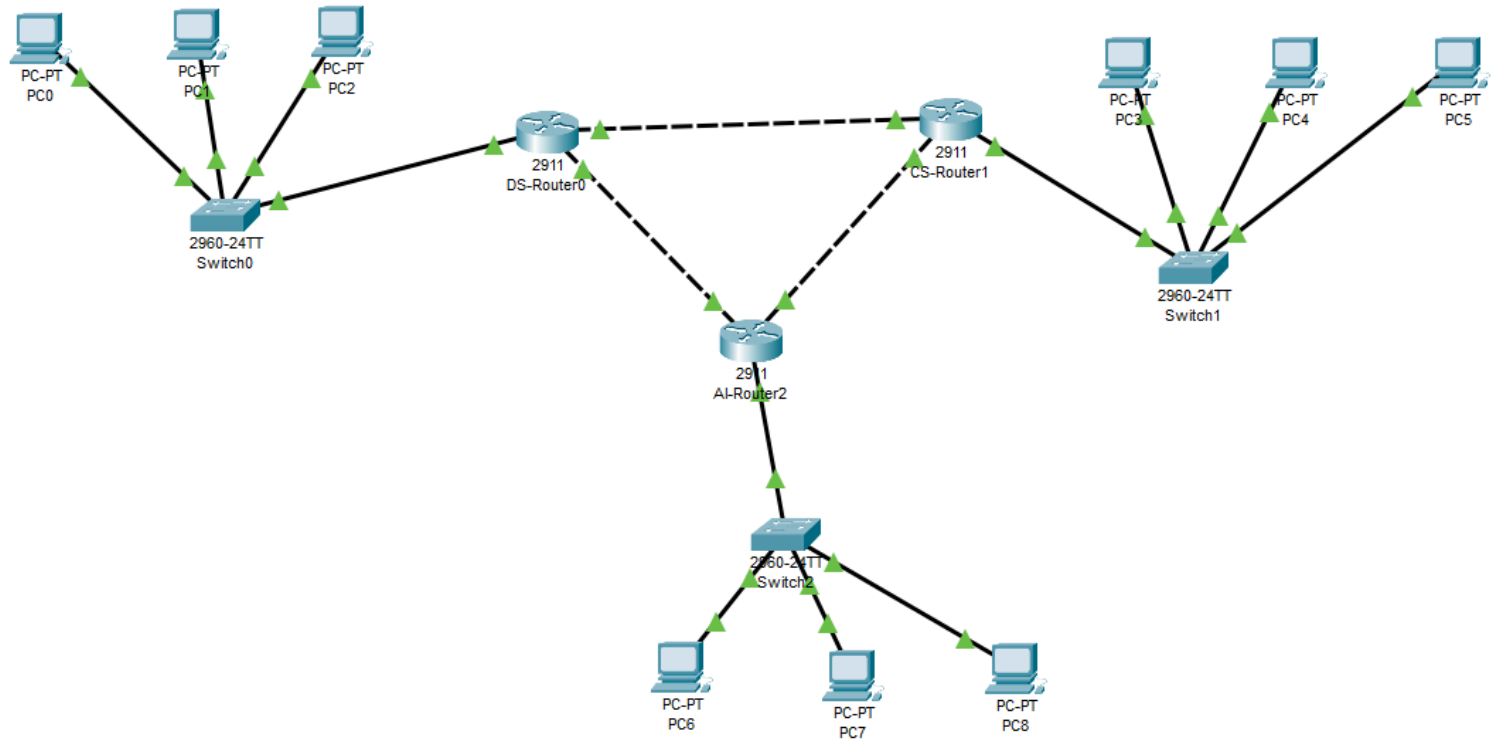
Host bits = 4 subnet mask = /28

Subnet address = 192.168.50.192/28

Range = 192.168.50.193 - 192.168.50.206

Broadcast address = 192.168.50.207

Q3:



DS-Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/1

Port Status

☒ On

Bandwidth

☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address0006.2AC5.7902

IP Configuration

IPv4 Address195.178.100.97

Subnet Mask255.255.255.252

Tx Ring Limit10

DS-Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/2

Port Status

☒ On

Bandwidth

☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address0006.2AC5.7903

IP Configuration

IPv4 Address195.178.100.105

Subnet Mask255.255.255.252

Tx Ring Limit10

CS-Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status

☒ On

Bandwidth

☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address00D0.BC87.E501

IP Configuration

IPv4 Address195.178.100.33

Subnet Mask255.255.255.224

Tx Ring Limit10

CS-Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/1

Port Status

☒ On

Bandwidth

☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address00D0.BC87.E502

IP Configuration

IPv4 Address195.178.100.101

Subnet Mask255.255.255.252

Tx Ring Limit10

CS-Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/2

Port Status

☒ On

Bandwidth

☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address00D0.BC87.E503

IP Configuration

IPv4 Address195.178.100.106

Subnet Mask255.255.255.252

Tx Ring Limit10

AI-Router2

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status

☒ On

Bandwidth

☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address00D0.97E3.7901

IP Configuration

IPv4 Address195.178.100.65

Subnet Mask255.255.255.224

Tx Ring Limit10

AI-Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.97E3.7902

IP Configuration

IPv4 Address 195.178.100.98

Subnet Mask 255.255.255.252

Tx Ring Limit 10

AI-Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/2

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.97E3.7903

IP Configuration

IPv4 Address 195.178.100.102

Subnet Mask 255.255.255.252

Tx Ring Limit 10

DS-Router0

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    195.178.100.0/24 is variably subnetted, 6 subnets, 3 masks
C       195.178.100.0/27 is directly connected, GigabitEthernet0/0
L       195.178.100.1/32 is directly connected, GigabitEthernet0/0
C       195.178.100.96/30 is directly connected, GigabitEthernet0/1
L       195.178.100.97/32 is directly connected, GigabitEthernet0/1
C       195.178.100.104/30 is directly connected, GigabitEthernet0/2
L       195.178.100.105/32 is directly connected, GigabitEthernet0/2

Router#
```

AI-Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

195.178.100.0/24 is variably subnetted, 6 subnets, 3 masks
C       195.178.100.64/27 is directly connected, GigabitEthernet0/0
L       195.178.100.65/32 is directly connected, GigabitEthernet0/0
C       195.178.100.96/30 is directly connected, GigabitEthernet0/1
L       195.178.100.98/32 is directly connected, GigabitEthernet0/1
C       195.178.100.100/30 is directly connected, GigabitEthernet0/2
L       195.178.100.102/32 is directly connected, GigabitEthernet0/2

Router#
```

CS-Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

195.178.100.0/24 is variably subnetted, 6 subnets, 3 masks
C       195.178.100.32/27 is directly connected, GigabitEthernet0/0
L       195.178.100.33/32 is directly connected, GigabitEthernet0/0
C       195.178.100.100/30 is directly connected, GigabitEthernet0/1
L       195.178.100.101/32 is directly connected, GigabitEthernet0/1
C       195.178.100.104/30 is directly connected, GigabitEthernet0/2
L       195.178.100.106/32 is directly connected, GigabitEthernet0/2

Router#
```

DS-Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

RIP Routing (v2)

Network

Add

Network Address

195.178.100.0

CS-Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

RIP Routing (v2)

Network

Add

Network Address

195.178.100.0

AI-Router2

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

RIP Routing (v2)

Network

Add

Network Address

195.178.100.0